

Test Cases

Case 1:

Input Tokens

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$$

Expected Self-Attention Output

5	6	7	8
5	6	7	8

Python Output

```
===== Q MATRIX =====
[[1 2 3 4]
 [5 6 7 8]]

===== K MATRIX =====
[[1 2 3 4]
 [5 6 7 8]]

===== V MATRIX =====
[[1 2 3 4]
 [5 6 7 8]]

===== RAW SCORES =====
[[ 30  70]
 [ 70 174]]

===== SCALED SCORES =====
[[15 35]
 [35 87]]

===== ATTENTION WEIGHTS (Q8.8 integers, 256 = 1.0) =====
[[  0 256]
 [  0 256]]

As fractions:
Row 0: ['0.0%', '100.0%'] (sum = 1.0000)
Row 1: ['0.0%', '100.0%'] (sum = 1.0000)

===== FINAL OUTPUT =====
[[5 6 7 8]
 [5 6 7 8]]
PS D:\attention-hdl-project>
```

HDL/Hardware Output

```
===== FINAL OUTPUT =====
OUT[0][0] = 5
OUT[0][1] = 6
OUT[0][2] = 7
OUT[0][3] = 8
OUT[1][0] = 5
OUT[1][1] = 6
OUT[1][2] = 7
OUT[1][3] = 8
```

GTKWave Signal Output



Case 2:

Input Tokens

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \end{bmatrix}$$

Expected Self-Attention Output

1.88	1.88	1.88	1.88
1.98	1.98	1.98	1.98

Python Output

```
===== Q MATRIX =====
[[1 1 1 1]
 [2 2 2 2]]

===== K MATRIX =====
[[1 1 1 1]
 [2 2 2 2]]

===== V MATRIX =====
[[1 1 1 1]
 [2 2 2 2]]

===== RAW SCORES =====
[[ 4  8]
 [ 8 16]]

===== SCALED SCORES =====
[[2 4]
 [4 8]]

===== ATTENTION WEIGHTS (Q8.8 integers, 256 = 1.0) =====
[[ 30 225]
 [  4 251]]

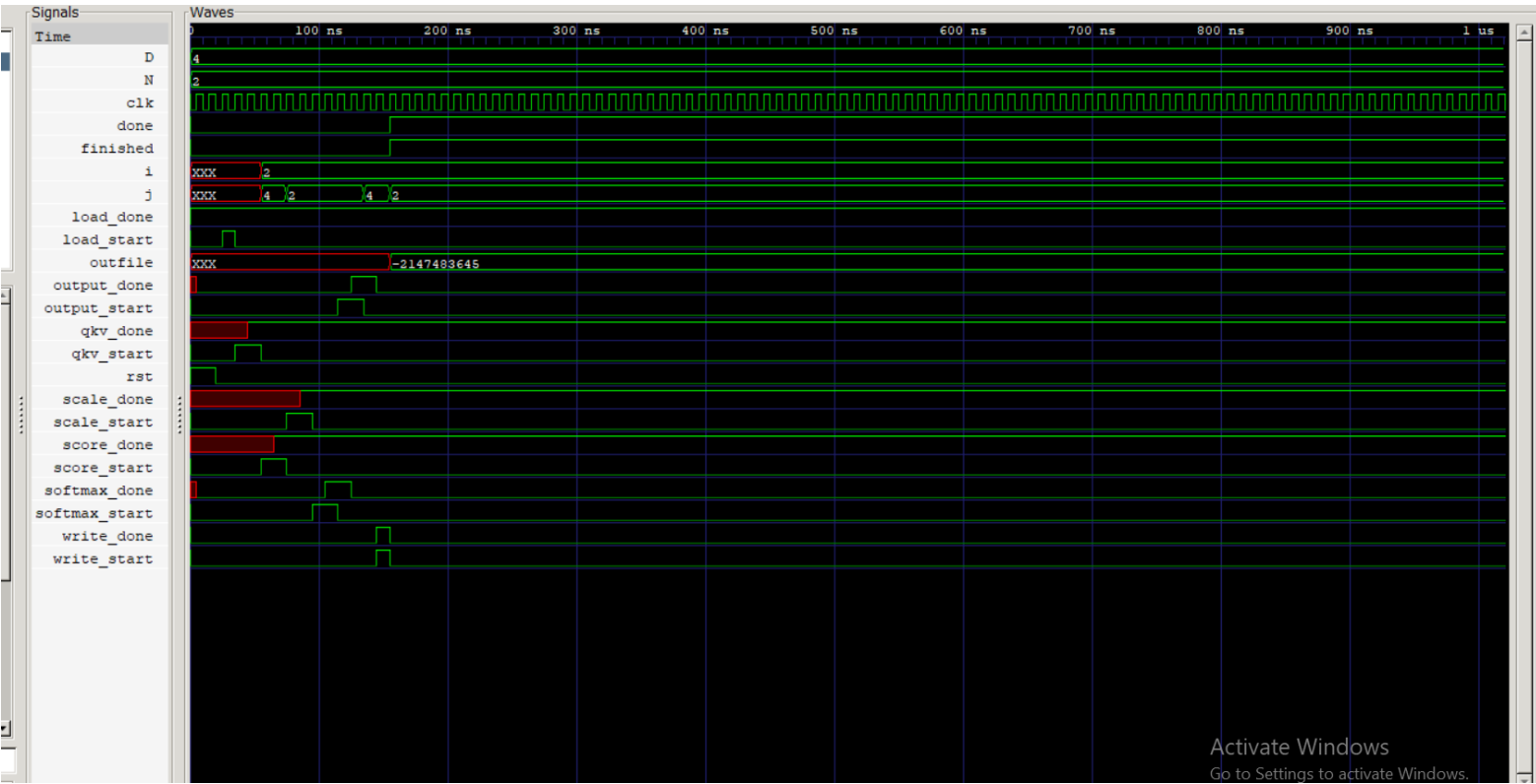
As fractions:
Row 0: ['11.7%', '87.9%'] (sum = 0.9961)
Row 1: ['1.6%', '98.0%'] (sum = 0.9961)

===== FINAL OUTPUT =====
[[1 1 1 1]
 [1 1 1 1]]
```

HDL/Hardware Output

```
===== FINAL OUTPUT =====
OUT[0][0] = 1
OUT[0][1] = 1
OUT[0][2] = 1
OUT[0][3] = 1
OUT[1][0] = 1
OUT[1][1] = 1
OUT[1][2] = 1
OUT[1][3] = 1
```

GTKWave Signal Output



Case 3:

Input Tokens

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Expected Self-Attention Output

$$\begin{bmatrix} 0.62 & 0.38 & 0 & 0 \\ 0.38 & 0.62 & 0 & 0 \end{bmatrix}$$

Python Output

```
===== Q MATRIX =====
[[1 0 0 0]
 [0 1 0 0]]

===== K MATRIX =====
[[1 0 0 0]
 [0 1 0 0]]

===== V MATRIX =====
[[1 0 0 0]
 [0 1 0 0]]

===== RAW SCORES =====
[[1 0]
 [0 1]]

===== SCALED SCORES =====
[[0 0]
 [0 0]]

===== ATTENTION WEIGHTS (Q8.8 integers, 256 = 1.0) =====
[[128 128]
 [128 128]]

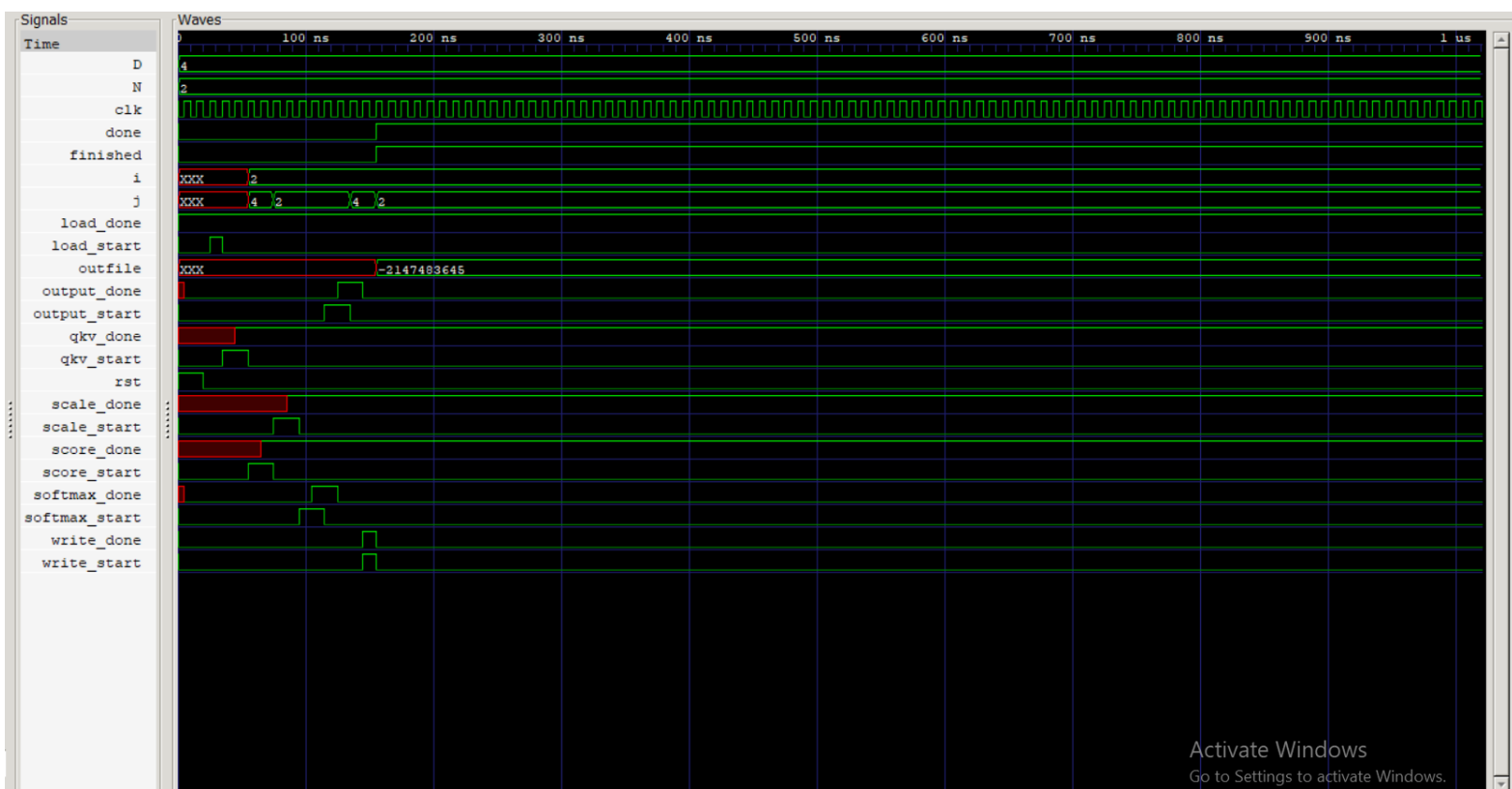
As fractions:
Row 0: ['50.0%', '50.0%'] (sum = 1.0000)
Row 1: ['50.0%', '50.0%'] (sum = 1.0000)

===== FINAL OUTPUT =====
[[0 0 0 0]
 [0 0 0 0]]
PS D:\attention-hdl-project> |
```

HDL/Hardware Output

```
===== FINAL OUTPUT =====
OUT[0][0] = 0
OUT[0][1] = 0
OUT[0][2] = 0
OUT[0][3] = 0
OUT[1][0] = 0
OUT[1][1] = 0
OUT[1][2] = 0
OUT[1][3] = 0
```

GTKWave Signal Output



Case 4:

Input Tokens

$$\begin{bmatrix} 3 & 1 & 2 & 0 \\ 1 & 0 & 1 & 2 \end{bmatrix}$$

Expected Self-Attention Output

$$\begin{bmatrix} 2.96 & 0.99 & 1.99 & 0.03 \\ 1.36 & 0.18 & 1.18 & 1.64 \end{bmatrix}$$

Python Output

```
===== Q MATRIX =====
[[3 1 2 0]
 [1 0 1 2]]

===== K MATRIX =====
[[3 1 2 0]
 [1 0 1 2]]

===== V MATRIX =====
[[3 1 2 0]
 [1 0 1 2]]

===== RAW SCORES =====
[[14  5]
 [ 5  6]]

===== SCALED SCORES =====
[[7 2]
 [2 3]]

===== ATTENTION WEIGHTS (Q8.8 integers, 256 = 1.0) =====
[[254  1]
 [ 68 187]]

As fractions:
Row 0: ['99.2%', '0.4%'] (sum = 0.9961)
Row 1: ['26.6%', '73.0%'] (sum = 0.9961)

===== FINAL OUTPUT =====
[[2 0 1 0]
 [1 0 1 1]]
PS D:\attention-hdl-project>
```

HDL/Hardware Output

```
===== FINAL OUTPUT =====
OUT[0][0] = 2
OUT[0][1] = 0
OUT[0][2] = 1
OUT[0][3] = 0
OUT[1][0] = 1
OUT[1][1] = 0
OUT[1][2] = 1
OUT[1][3] = 1
```

GTKWave Signal Output

